

INFO6007

Project Management in IT

Week 11: IT Service Management and processes 1

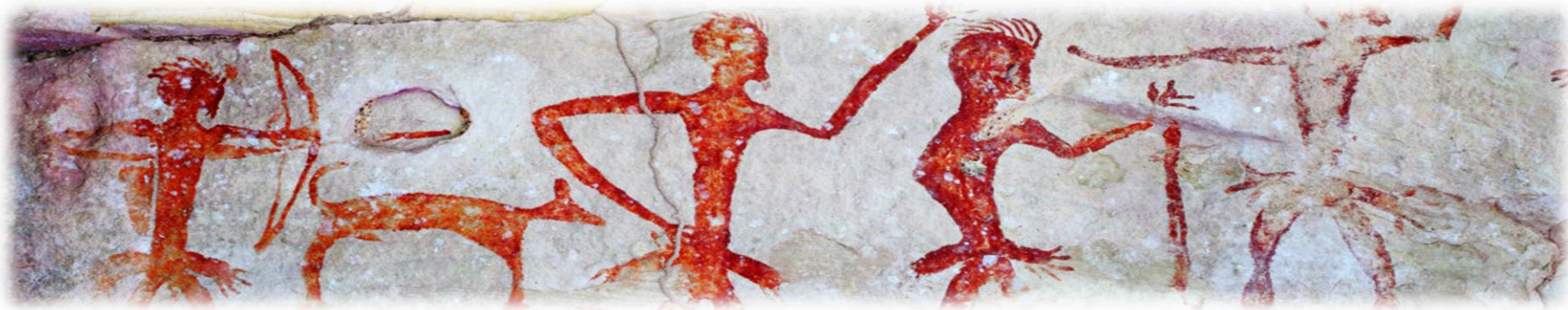
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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Agenda

- Introduction
- Service Relationship Models
- ITIL v4.0

Introduction

What is ITSM (IT Service Management)?

- It is defined as a **set of policies, processes, and practices** that organisations use to **design, deliver, manage, and improve IT services** to meet the needs of the business and its customers.
- Key concepts:
 - Service oriented approach
 - Processes and practices: ITSM relies on **structured processes** like **incident management, problem management, change management, service level management**.
 - Value creation: ITSM is **business-aligned**: IT services exist to **enable business outcomes**, not just to operate technology.
 - Frameworks and Standards: ITSM is implemented using **best-practice frameworks** such as ITIL, COBIT, etc.

Why ITSM matters?

- **Business Alignment:** Ensures IT supports organisational goals.
- **Improved Service Quality:** Reduces downtime and service failures.
- **Cost Efficiency:** Optimises resources and reduces duplication.
- **Customer Satisfaction:** Provides reliable, predictable IT services.
- **Continuous Improvement:** Iterative improvements based on feedback and metrics.

Services in IT

- In IT, a service is a means of **delivering value** to customers by **facilitating desired outcomes** - without the customer having to own or manage the underlying technology, costs, or risks.
- Formal definition by ITIL V4.0 is:
 - A service is a means of enabling **value co-creation** by facilitating **outcomes** that customers want to achieve, without the customer having to manage specific costs and risks.

Where is ITSM Used?

- **Education**
 - LMS, student portals, Wi-Fi, and IT help desks
- **Finance**
 - Online banking, ATM systems, cybersecurity, and incident response
- **Healthcare**
 - Patient records, appointment systems, and clinical IT support
- **Government**
 - Citizen portals, internal systems, compliance, and security
- **Business and Industry**
 - ERP systems, cloud platforms, service desks, and operational IT support

Services in IT

- Key characteristics of IT Services:

Characteristic	Explanation	Example
Intangible	Services can't be touched or stored; they're experienced.	Cloud hosting, email access
Outcome focused	The goal is what the customer can do, not the technology used.	Streaming movies (Netflix)
Co-created value	Both provider and consumer contribute to the outcome.	University IT + Lecturer = Online teaching
Continuous Delivery	Services are ongoing, not one-off products.	IT helpdesk, cybersecurity monitoring
Risk and Cost transfer	Provider handles complexity and risk.	Using AWS instead of maintaining your own data centre

Services in IT

- Types of IT Services
 - Infrastructure Services
 - Provide the foundational hardware, software, and networking.
 - Examples: Cloud hosting (AWS EC2), data storage, network management.
 - Application Services
 - Enable users to perform specific business tasks.
 - Examples: Microsoft 365, Canvas LMS, Salesforce CRM.
 - Business Services
 - Integrate IT services to deliver end-to-end business value.
 - Examples: Payroll processing, online banking, e-commerce operations.
 - Support Services
 - Maintain and improve the quality of IT operations.
 - Examples: IT Helpdesk, Service Desk, Technical Support.

Products in IT

- A **tangible or intangible deliverable** that provides specific functionality or features, created, developed, and owned by a business to **generate value through sale, licensing, or use**.

Characteristic	Explanation	Example
Tangible/Defined Output	The product is a complete entity, software, or hardware.	A mobile app, laptop, or ERP system
Ownership	Once purchased, the customer owns or licenses it.	Buying Windows OS or Photoshop
Feature based value	The value lies in functionality and performance.	Excel's data analysis tools
One time or Subscription sale	Sold once or as a renewable license.	Microsoft Office (one-time) vs 365 (subscription)
Maintenance Optional	Updates and support may be separate.	Paying extra for software updates or warranty
Stable and Versioned	Released in versions or editions.	iPhone 16, Windows 11, or Ubuntu 24.04

Products in IT: Types of IT Products

- Hardware Products: Physical computing devices.
 - Examples: Laptops, routers, servers, IoT devices.
- Software Products: Packaged or downloadable applications.
 - Examples: Microsoft Word, Adobe Photoshop, AutoCAD.
- Digital Products: Intangible products distributed online.
 - Examples: eBooks, digital music, NFTs, mobile apps.
- Platform Products: Serve as a foundation for services or ecosystems.
 - Examples: iOS, Android, AWS platform, Salesforce.
- Data Products: Data sets, APIs, or analytical models offered as a product.
 - Examples: Google Maps API, financial data feeds, AI models.

Services vs Products in IT

Aspect	Product	Services
Nature	Tangible or defined software/hardware	Intangible and experience-based
Ownership	Customer owns or licenses	Customer uses, provider owns
Delivery	One-time or versioned	Continuous
Value Focus	Features and functions	Outcomes and experiences
Example	Buying Adobe Photoshop	Subscribing to Adobe Creative Cloud

Why does ITSM matter?

- **IT Service Management (ITSM)** matters because it ensures that IT is **managed strategically, efficiently, and consistently** delivering **reliable, high-quality services** that directly support **business goals and customer satisfaction**.



(Heimdal, 2025)

Benefits of ITSM

- Alignment of IT with Business Goals
 - ITSM ensures IT services are designed and delivered based on **business needs**, not just technology trends.
 - IT activities are **linked to strategic objectives** and **measurable outcomes**.
- Consistency and Standardisation
 - ITSM introduces **structured processes** (incident, problem, change, release, and service request management).
 - Ensures IT operations are **predictable**, **auditable**, and **repeatable** across departments.
- Cost optimisation and Efficiency
 - ITSM helps organisations understand **where resources are used**, eliminate waste, and improve productivity.
 - Through frameworks like ITIL, it enables **better prioritisation**, **automation**, and **resource management**.

Benefits of ITSM

- Enhanced customer and user experience
 - ITSM focuses on **value delivery** and **service quality** from the **end-user's perspective**.
 - Service Level Agreements (SLAs) define expectations for uptime, response, and resolution times.
- Continuous improvement and agility
 - ITSM promotes **feedback loops**, **metrics**, and **continuous improvement** through the *Plan–Do–Check–Act (PDCA)* cycle.
 - Enables organisations to **adapt quickly to technological or business change**.
- Risk management and governance
 - ITSM defines **controls**, **documentation**, and **accountability**, supporting compliance with laws and standards (e.g., ISO 20000, GDPR, NIST).
 - Helps mitigate risks related to security, availability, and service disruptions.

IT Project Management vs IT Service Management

Aspect	IT Project Management	IT Service Management
Definition	The discipline of planning, executing, and closing IT-related projects within scope, time, and budget.	The discipline of designing, delivering, managing, and improving IT services that support business operations.
Objective	To deliver a unique product, service, or result successfully.	To ensure ongoing, reliable delivery of IT services to customers and users.
Focus	Creating or changing something (e.g., developing software, upgrading infrastructure).	Maintaining and optimizing what's already running (e.g., managing incidents, changes, and SLAs).
Primary Stakeholders	Project sponsor, project manager, development teams, business users.	Service manager, service desk, end users, customers.
Typical Roles	Project Manager, Scrum Master, Developers, QA Team.	Service Manager, Incident Manager, Change Manager, Service Desk.

Menti: Differentiate PM and ITSM using one sentence

- <https://www.menti.com/aluypqppmo9z>



Facilitating role of services in an Economy

- In a modern economy, **services are the primary enablers of value creation.**
- Services play a **facilitating role** by enabling other sectors to operate efficiently and innovate effectively. They act as the “**glue**” that holds the economy together, ensuring smooth functioning and connectivity between businesses, governments, and consumers.
- In the **Information Age** services, particularly **IT services**, enable:
 - **Digital transformation** of traditional industries.
 - **Automation and data-driven decision-making.**
 - **Global collaboration** through cloud and communication platforms.
 - **Scalability and flexibility** for businesses via “as-a-service” models (SaaS, IaaS, PaaS).



Service Relationship Models

Introduction

- Service relationship models describe **how services are exchanged** between **providers** and **consumers** in an economy.
- They define **who provides the service, who receives it, and how value is co-created** between them.

Business Oriented

- Business-2-Business
 - **Definition:** Services exchanged **between two businesses or organisations.**
 - **Purpose:** To support business operations, efficiency, or profitability.
 - B2B services focus on efficiency, scalability, and reliability.
 - *Examples: AWS, Salesforce, Intel, Accenture*
- Business-2-Consumer
 - **Definition:** Services delivered **directly from a business to an individual consumer.**
 - **Purpose:** To meet personal or end-user needs.
 - B2C services emphasize user experience, personalization, and convenience.
 - *Examples: Netflix, Apple, Uber, Spotify*

Business Oriented

- Business-2-Government
 - **Definition:** Services provided by **private companies to government agencies or departments.**
 - **Purpose:** To enable or enhance **public administration and citizen services.**
 - B2G focuses on compliance, transparency, and long-term contracts.
 - *Examples: IBM, Thales Group, Serco, Telstra*

Customer Oriented

- Consumer-2-Consumer
 - It refers to transactions or services where **individual consumers exchange goods, services, or information directly** with other consumers often facilitated by a **third-party digital platform**.
 - In IT, C2C models often rely on **secure digital infrastructure, data analytics, and trust mechanisms** like reputation scores, payment gateways, and dispute resolution systems.
 - *Examples: eBay, Facebook Marketplace, Gumtree, Airbnb*
- Consumer-2-Business
 - **C2B** flips the traditional model, individual **consumers offer value to businesses** by providing products, services, or data that businesses use to enhance their operations or marketing.
 - In IT and digital ecosystems, Consumers provide **data, feedback, content, or development work** to help businesses improve products or services. AI and analytics use consumer-generated data to refine systems.
 - *Examples: YouTube (content creators earning from ads), Shutterstock (individuals selling photos), Fiverr*

Government Oriented

- Government-2-Business
 - **G2B** refers to the relationship where **government agencies provide services, information, or regulatory support to businesses.**
 - These services help companies **comply with laws, access resources, or participate in public procurement.**
 - *Examples: Business.gov.au, Australian Taxation Office (ATO) Business Portal, eProcurement systems, ASIC online services*
- Government-2-Consumer
 - **G2C** represents services delivered **by government agencies directly to individuals (citizens or residents).**
 - These services aim to **enhance public welfare, accessibility, and engagement.**
 - *Examples: MyGov, Service NSW, Medicare Online, Australian Passport Application Portal*

Government Oriented

- Government-2-Government
 - **G2G** refers to service relationships and collaborations **between different government entities**, either **within a single level of government** (e.g., local-to-state) or **across levels** (e.g., federal-to-state).
 - *Examples: ABS–Home Affairs data sharing, GovLink, UN e-Government portal, Interpol information systems*

Upcoming trends in Service Relationship Models

- B2E
 - **B2E** refers to services that a **business provides directly to its employees**, typically through internal IT systems, HR platforms, or enterprise applications.
 - The goal is to **enhance employee productivity, engagement, and satisfaction**.
 - *Examples: Microsoft Teams, Workday, SAP SuccessFactors, Google Workspace*
- D2C
 - **D2C** refers to when **manufacturers or service providers sell directly to end consumers**, bypassing intermediaries like distributors or retailers.
 - Enabled primarily by **digital platforms and e-commerce** technologies.
 - *Examples: Nike, Tesla, Warby Parker, Glossier*
- B2M
 - **B2M** (Business-to-Many) describes when a business **serves multiple types of customers**, such as **both other businesses (B2B) and consumers (B2C)**, using a **hybrid model**.
 - *Examples: Amazon, Microsoft, Google, Adobe*

Upcoming trends in Service Relationship Models

- M2M
 - **Machine-to-Machine (M2M)** refers to **direct communication between devices** without human intervention, typically over wired or wireless networks.
 - This model forms the **foundation of IoT (Internet of Things)**.
 - *Examples: Smart meters, connected cars, industrial sensors, IoT home devices (e.g., Nest Thermostat)*
- N2C
 - **Network-to-Consumer (N2C)** describes a model where **a network or platform**, rather than a single company, **provides services directly to consumers**.
 - It's a hallmark of the **platform economy** and **digital ecosystems**.
 - *Examples: Google Play Store, Apple App Store*
- N2B
 - **Network-to-Business (N2B)** describes services delivered from **a networked platform or ecosystem** to **businesses**.
 - Instead of a single provider, multiple connected service providers operate collaboratively to offer business solutions.
 - *Examples: Salesforce AppExchange, AWS Marketplace, Google Ads Network*

The competitive role of Information

- Information is not merely a support tool in modern organisations, it is a **strategic resource** that can create, sustain, and protect **competitive advantage**.
- In the digital era, the **effective collection, management, and use of information** determine how successfully a company can deliver value through its IT services and products.

Dimension	Role of Information	Outcome
Operational	Streamlines processes and reduces costs	Efficiency
Strategic	Guides competitive positioning	Advantage
Innovative	Enables new products and services	Differentiation
Managerial	Supports data-driven decisions	Accuracy
Customer-focused	Enhances user experience	Loyalty
Governance	Ensures ethical, secure use	Trust

ITIL v4.0

The 4 dimensions of Service Management

- The **Four Dimensions of Service Management** provide a **holistic view** of the factors that collectively ensure **effective and efficient service delivery**.
- These dimensions are **interdependent**, meaning that neglecting one can weaken or destabilize the entire **Service Value System (SVS)**.

Dimension	Focus Area	Key Question
Organisation and People	Structure, culture, roles, and skills	Do we have the right people and culture to deliver value?
Information and Technology	Data, tools, and digital systems	Are our technologies enabling effective service delivery?
Partners and Suppliers	External relationships and contracts	Who helps us co-create and deliver value?
Value Streams and Processes	Activities that create value	How do we transform inputs into valuable outcomes?

The 4 dimensions of Service Management

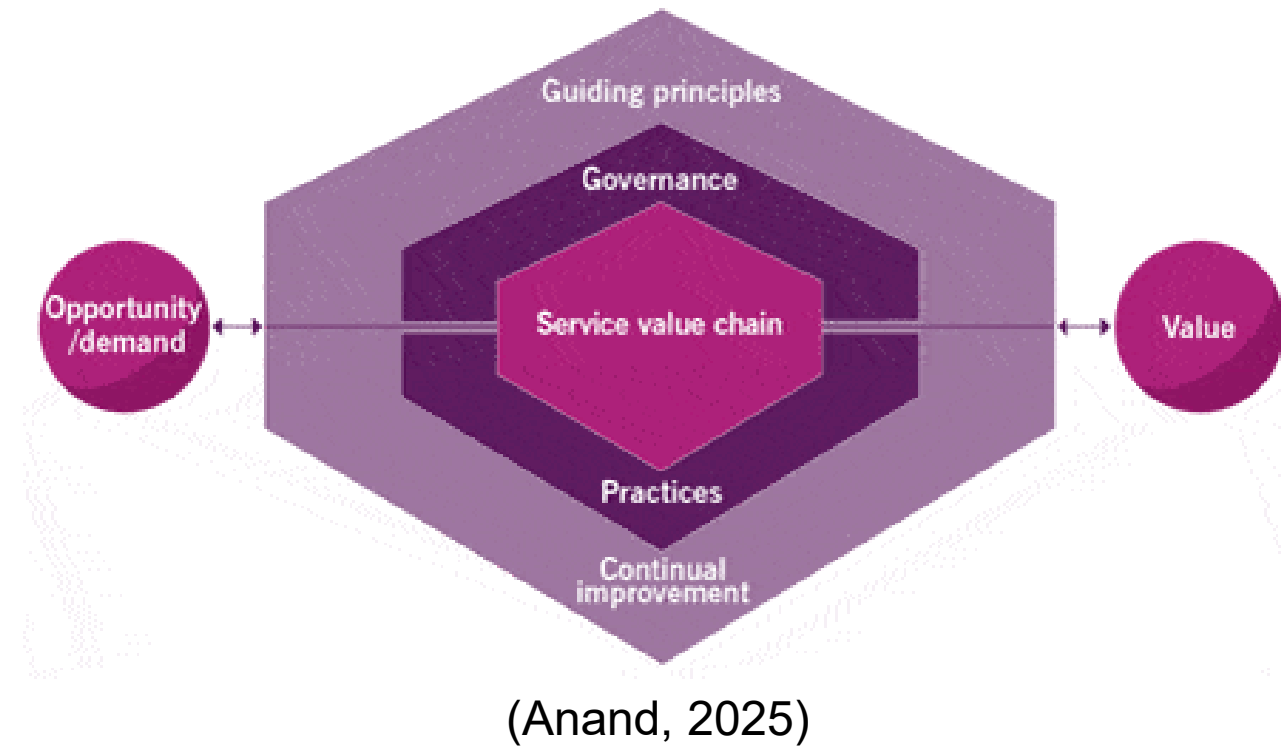
- Organisations and People
 - This dimension ensures that the organisation's **culture, structure, competencies, and communication** support its overall **service strategy** and **value creation**. Without the right people or culture, even the best technology or processes will fail.
 - ITIL stresses **human-centered service delivery**, balancing technical efficiency with empathy and communication.
- Information and Technology
 - This dimension covers the **information, knowledge, and technologies** required for managing and delivering services. It includes both **technical infrastructure** and **the data required for decision-making**.
 - Information and technology enable the **automation and digitalization** of services.
 - They also support decision-making by providing insights into service performance, customer satisfaction, and process efficiency.

The 4 dimensions of Service Management

- Partners and Suppliers
 - This dimension focuses on the **third parties** (vendors, suppliers, service providers, partners) that the organisation relies on to co-create and deliver services.
 - Modern IT services rely heavily on **ecosystems of partners and suppliers**. Effective ITSM must ensure **clear communication, defined responsibilities**, and **aligned objectives** across all parties.
- Value Streams and Processes
 - This dimension describes **how value is created and delivered** through **coordinated processes and workflows**. It ensures that activities are **efficient, measurable, and outcome-focused**.
 - This dimension aligns with the **ITIL 4 Service Value Chain**, where services are co-created through **value streams** that combine people, technology, and partners.

ITIL Overview

- **ITIL** stands for **Information Technology Infrastructure Library**, a **globally recognised framework** that provides **best practices for delivering high-quality IT services**.
- It defines **how IT services should be aligned with business needs**, ensuring consistency, efficiency, and continual improvement.



Evolution to ITIL v4.0

Version	Year Released	Focus/Characteristics
ITIL v1.0	1980s-1990s	Collection of 30+ books covering IT operations.
ITIL v2.0	Early 2000s	Streamlined to focus on Service Support and Service Delivery.
ITIL v3.0	2007/2011	Introduced Service Lifecycle (5 stages) and value co-creation.
ITIL v4.0	2019	Modernised for digital transformation , Agile , DevOps , and cloud environments .

Key Components of ITIL v4.0

- Service Value System (SVS)
- Guiding Principles
- Governance
- Service Value Chain (SVC)
- ITIL 4 Practices
- Continual Improvement

1. Service Value System

- The **Service Value System (SVS)** in ITIL 4 provides a **holistic view of how all the components and activities of an organisation work together** to enable **value co-creation with stakeholders** through IT-enabled services.
- Components of the SVS:

Component	Role/purpose
Guiding Principles	Universal recommendations that guide decisions and actions. They ensure consistency and alignment with organisational values.
Governance	Ensures the organisation is directed and controlled, monitoring policies, compliance, risk, and performance.
Service Value Chain (SVC)	The central operating model that converts demand into value through six activities: Plan, Improve, Engage, Design & Transition, Obtain/Build, Deliver & Support.
Practices	34 management practices (general, service, technical) that provide guidance for performing tasks efficiently.
Continual Improvement	A recurring activity that ensures ongoing enhancement of services, practices, and capabilities.

2. Guiding Principles

- The **Guiding Principles** are **universal recommendations** that guide **decision-making and actions** across the entire organization.
- They ensure that all efforts **align with value creation, stakeholder needs, and business objectives**.
- There are 7 guiding principles:
 - Focus on value
 - Start where you are
 - Progress iteratively with feedback
 - Collaborate and Promote visibility
 - Think and Work Holistically
 - Keep it simple and practical
 - Optimise and automate

Guiding principles explained

- Focus on Value
 - Everything the organization does should deliver **value to customers, users, and stakeholders**.
 - Consider both tangible (financial) and intangible (experience, satisfaction) benefits.
- Start where you are
 - Assess **current resources, processes, and capabilities** before introducing new solutions.
 - Avoid discarding existing processes unnecessarily.
- Progress iteratively with feedback
 - Break work into **smaller, manageable steps** instead of attempting massive changes at once.
 - Use **feedback loops** to guide improvement.

3. Governance

- Collaborate and Promote Visibility
 - Encourage **cross-functional collaboration** and **transparent communication**.
 - Reduces silos and ensures shared understanding.
- Think and work holistically
 - Consider the **entire organization**, including all four dimensions:
 - People & Organisations, Information and technology, partners & suppliers, and Value streams and processes.
- Keep it simple and practical
 - Avoid unnecessary complexity and bureaucracy.
 - Focus on **practical solutions** that deliver results efficiently.
- Optimise and automate
 - Identify **what can be improved or automated** to increase efficiency, reliability, and consistency

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For You!

End of Lecture Questions

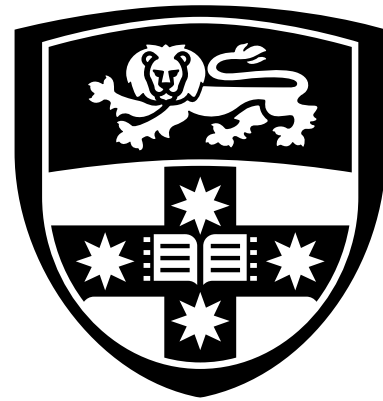
- Define **IT Service Management (ITSM)** and explain how it differs from IT project management.
- Explain why **ITSM matters in modern organizations**. How does it support business objectives?
- List and briefly describe the **Four Dimensions of Service Management** in ITIL 4.
- An organization struggles with **siloed IT teams** and poor communication. Which **guiding principles** could help address this, and how?
- Which of the **four dimensions of service management** is often the most neglected in organizations, and why?
- How can ITIL 4 principles be applied in a **digital transformation project** that includes cloud migration and AI-based service automation?

Case Study

- UniTech Services is the IT department of a large university experiencing rapid growth, including multiple campuses and extensive online learning programs. The department struggles with **fragmented IT processes, system outages, slow support responses, and limited service visibility**, leading to dissatisfaction among students and staff. To address these challenges, UniTech plans a **digital transformation initiative** using **ITIL 4-based IT Service Management (ITSM)**.
- Key initiatives include deploying a new **Learning Management System (LMS)**, automating incident and service request management, integrating **cloud services** from multiple vendors, and establishing a **central Service Desk**. The department also aims to adopt **Agile and DevOps practices** to improve collaboration, responsiveness, and continual service improvement.
- The goal is to **align IT services with stakeholder needs**, improve service reliability, enable value co-creation, and establish a framework for **ongoing monitoring, governance, and continual improvement** of all IT services.

Case-study based questions

- Describe how **partners and suppliers** (cloud vendors) can be managed to ensure reliable LMS operation.
- Suggest ways **information and technology** can be leveraged to improve service visibility and user satisfaction.
- Compare IT project management and IT service management approaches in the context of deploying the LMS. Which aspects of each approach should be prioritized?
- Propose a **high-level continual improvement plan** for the IT department over the next 12 months, integrating metrics for **incident management, SLA compliance, and user satisfaction**.



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